Ivor Ho

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Education

Bachelor of Science in Biology (Computational Biology) Expected May 2027

Florida State University, Tallahassee, FL 3.677 GPA

Research Experience

Undergraduate Research Fellow (June 2026 - August 2026)

Strand Lab, *University of Texas Southwestern* SURF, Dallas, Texas

- Integrated scRNA-seq, snRNA-seq, and snATAC-seq from human seminal vesicle and vas deferens using Seurat, annotating cell populations across both organs
- Distinguished genuine cell populations from low-information artifacts through iterative marker review, resolving ambiguous clusters with PI
- Selected markers via differential expression to design a Xenium Prime 5K panel, validating cell-type markers and spatial location in situ

CPRIT-CURE Intern (June 2025 - August 2025)

Ken Chen Lab, *MD Anderson Cancer Center* CPRIT CURE Program, Houston, Texas

- Extracted molecular interactions from curated databases (DoRothEA, CellPhoneDB, STRING, KEGG) into a directed knowledge graph using Python and R
- Applied personalized PageRank over the graph to predict which genes respond to a perturbation
- Designed randomized-graph controls to test prediction performance against a null baseline

SMART Intern (June 2024 - August 2024)

Shaulsky Lab, *Baylor College of Medicine* SMART Program, Houston, Texas

- Constructed chromoprotein reporter plasmids via GoldenBraid assembly and standard cloning (plasmid prep, restriction digestion)
- Introduced constructs by electroporation and maintained bacterial and *Dictyostelium* cultures
- Assayed reporter expression across developmental stages to test differentiation-state readouts

Undergraduate Research Assistant (August 2023 - Present)

Lemmon Lab, *Florida State University* UROP, Tallahassee, Florida

- Modeled the neural circuit underlying mate-choice behavior in chorus frogs using MATLAB and R
- Generated synthetic datasets and mapped call-parameter solutions across populations
- Applied multidimensional scaling to compare parameter solutions and interpret population-level structure

Presentations

Ho I., Paterson-Rees E., Malewska A., Lafin J., & Strand D. (2026). *A Single Cell and Spatial Atlas of the Human Seminal Vesicle and Vas Deferens*. [Poster Presentation] UTSW SURF Poster Symposium, Dallas, TX

Ho I., He S., Mohanty V., & Chen K. (2025). *Integrating molecular signature databases to holistically reveal gene signaling events*. [Poster Presentation] MD Anderson CPRIT CURE Poster Symposium, Houston, TX

Ho I., Kurasawa M., & Shaulsky G. (2024). *Exploring the expression of chromoproteins in Dictyostelium discoideum*. [Oral Presentation] Baylor College of Medicine SMART Program, Houston, TX

Weintraub C., Ho I., & Lemmon A. (2024). *Modeling Neural Circuits to Understand Reproductive Isolation and Cryptic Evolution*. [Poster Presentation] Florida State University Department of Scientific Computing Exposition (CX24), Tallahassee, FL

Ho I., & Lemmon A. (2024). *Modeling Neural Circuits to Understand Incipient Speciation Part 2: Quantifying the Potential for Cryptic Evolution*. [Poster Presentation] Florida State University Undergraduate Research Opportunity Program End of Year Poster Symposium, Tallahassee, FL

Honors/Awards

- FSU Dean's List 2023-2025
- FSU Honors Program Member 2023-present
- FSU UROP (Undergraduate Research Opportunity Program) 2023-2024

Community Service

Teacher

- Read English books and gave English presentations to elementary school students in Taiwan (May 2025)

Teaching Assistant

- Co-taught English with teachers in Taiwan to grade school students (May 2021 – June 2021)

Bluebonnet Classroom

- Taught English to international grade school students weekly via Zoom (October 2020 - July 2021)

Skills

Language: English, Chinese (Beginner spoken), Spanish (first level, Spoken)
Computational: MATLAB, R and RStudio, Python, Seurat
Project management: GitHub

Revised: August, 2026